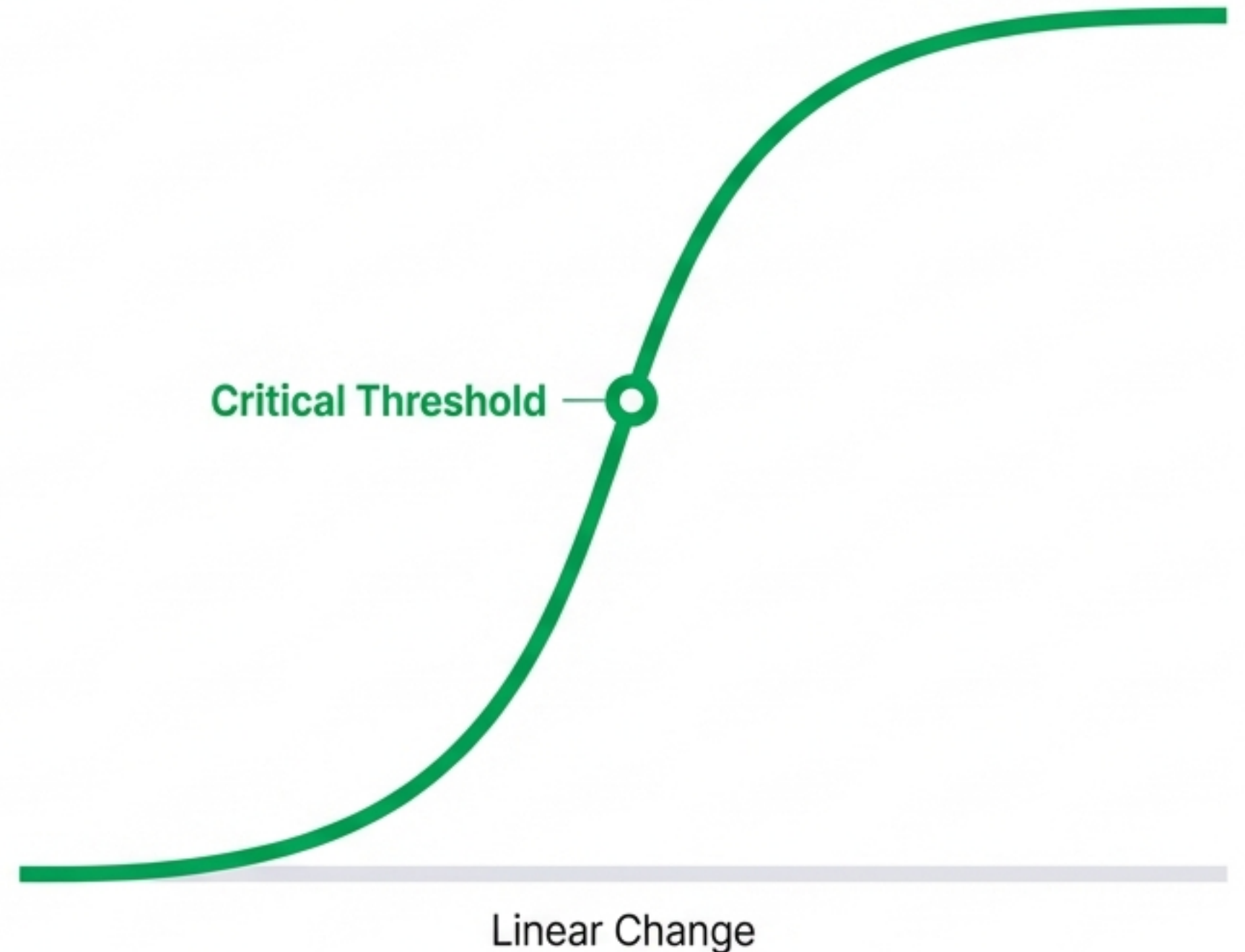


Green Tipping: Behavioural Shifts for a Sustainable Future

Moving beyond linear policy to trigger rapid, non-linear social transformation.

While technology and policy are critical, individual choices act as a powerful force in the sustainability transition. This deck, based on research by Professor Alessandro Tavoni and Veronica Pizziol, outlines the physics of "Social Tipping Points"—critical thresholds where small actions create self-reinforcing momentum.

Key Insight: We are exploring the shift from a "High-Carbon Status Quo" to a new societal state characterized by lower emissions.

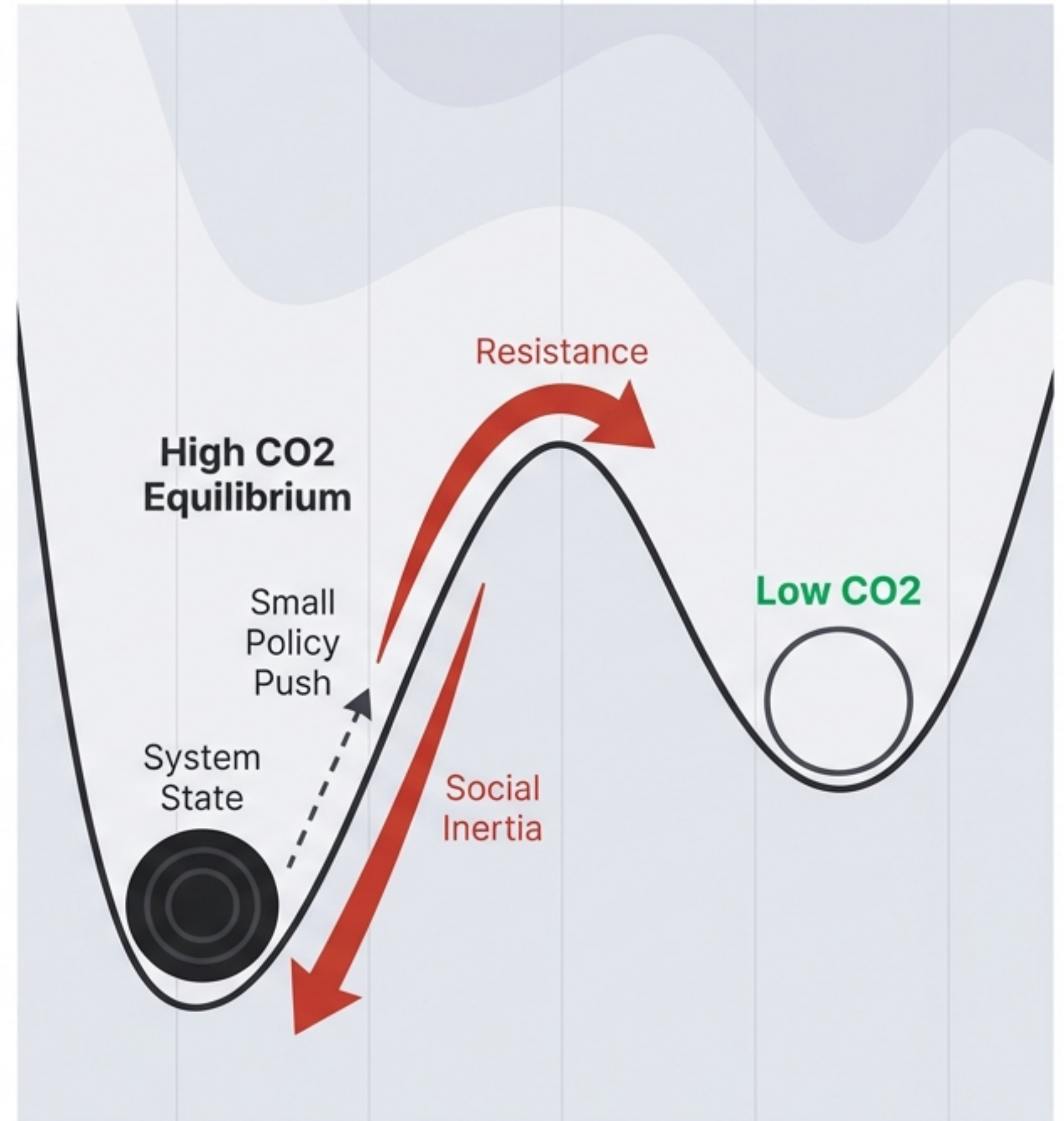


The Physics of Stagnation: Why the Status Quo Persists

Society functions as a system currently trapped in a “High-CO2 equilibrium”.

- **The Valley:** Represents the stability of current norms, infrastructure, and behaviours.
- **Resistance:** Small policy pushes move the “ball” slightly up the slope, but without sufficient force, gravity (social inertia) rolls it back to the bottom.
- **The Goal:** We cannot just push the ball; we must push it over the hill to the “Low CO2” valley.

Adapted from Pizziol and Tavoni (2024).



Defining Social Tipping Points

Social tipping points are critical thresholds in socioeconomic systems where small perturbations trigger significant, rapid, and difficult-to-reverse shifts in collective behaviours and norms.

Linear Change

Gradual, proportional response to effort.



(e.g. A 10% tax increase leads to a 2% drop in consumption).

Tipping Point

Non-linear, abrupt change driven by self-reinforcing contagion and feedback loops.
(e.g. Adoption explodes once a threshold is met).



Key Terminology

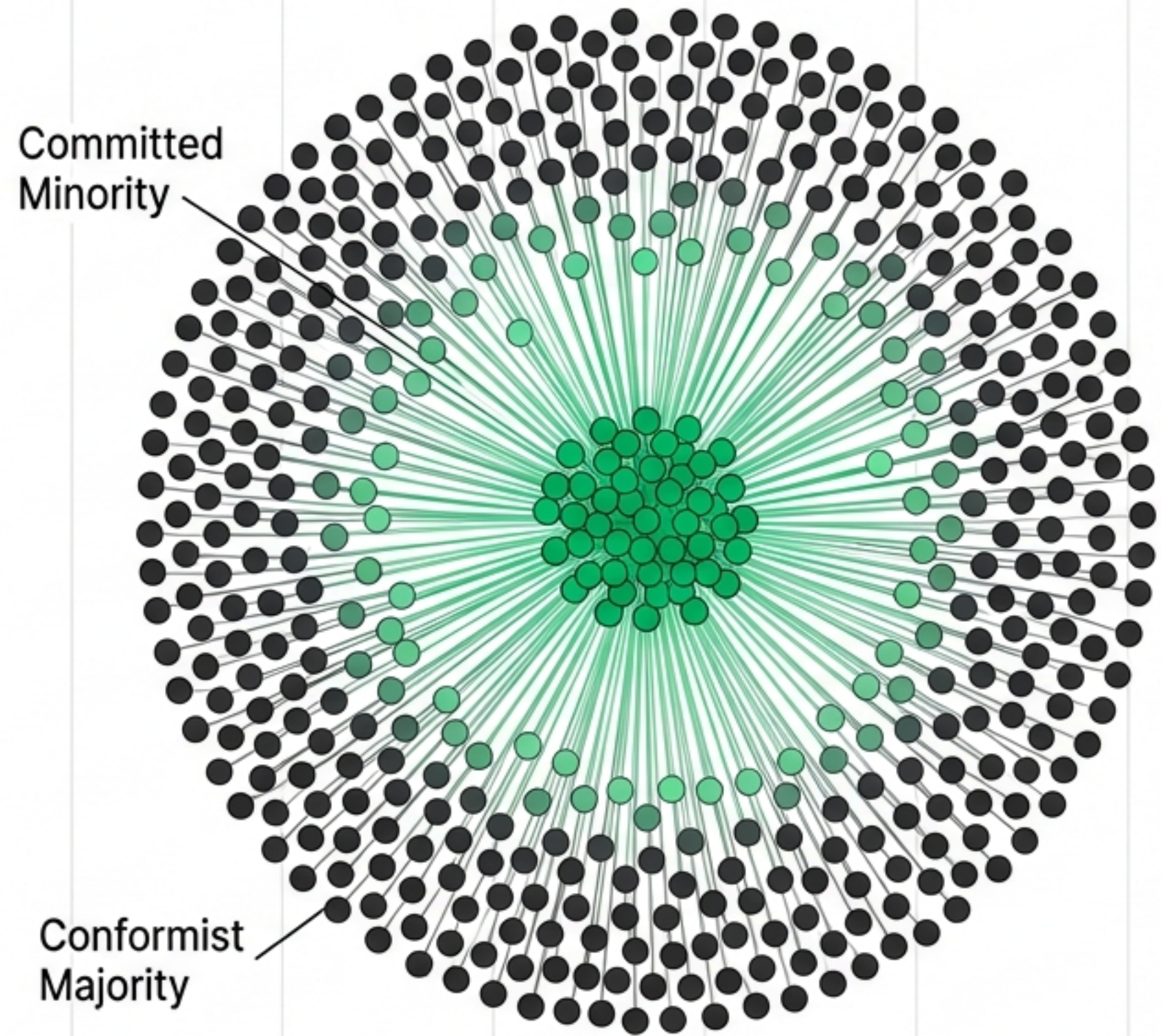
- **Self-Reinforcing Contagion:** The swift spread of new behaviours (e.g., from a niche to a norm).
- **Irreversibility:** Once the threshold is crossed, the new norm becomes the stable state.

The Critical Threshold: The Power of the Committed Minority

Tipping points are rarely triggered by the majority. They are ignited by “committed minorities”—trendsetters, change instigators, or early adopters—who refuse to follow entrenched norms.

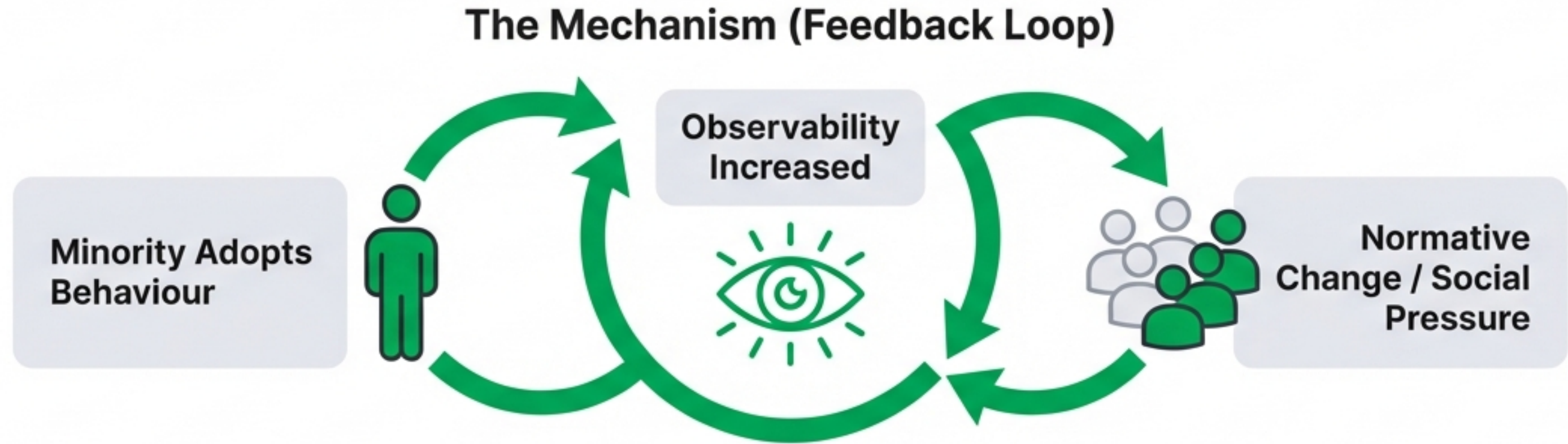
The “**Magic Number**”: Research suggests the critical threshold for a committed minority to sway the majority ranges between 10% and 40% of the population.

- **The Minority:** High intrinsic motivation; they demand change and push for alternative lifestyles.
- **The Majority:** Characterized as “conformists” with little motivation to change until the social cost of not changing becomes too high.



Mechanics of Contagion and Scale

The direct influence of a minority group is limited until it triggers a feedback loop. This happens through Normative Change (redefining what is acceptable) and Observability (making the behaviour visible).



Case Studies in Scale



Hush Puppies (The 1990s)

As noted by Malcolm Gladwell, a small group of trendsetters in NYC sparked a global cultural phenomenon, driving a massive resurgence in sales.



Fridays for Future

A prime example of social contagion. It began in August 2018 with a single participant, Greta Thunberg. Through visibility and resonance, it grew into a social movement of six million people within one year.

Tipping Points in the Wild: Sustainability Success Stories

Recent evidence confirms that tipping dynamics are actively shaping the green transition.



Solar Panels (USA)

Adoption of rooftop solar is driven by neighbourhood effects; seeing panels on a neighbour's roof significantly increases the likelihood of installation.



Urban Transport (Copenhagen)

The transition from cars to bicycles was not linear but a structural shift in norms and infrastructure.



Historical Precedent: Smoking

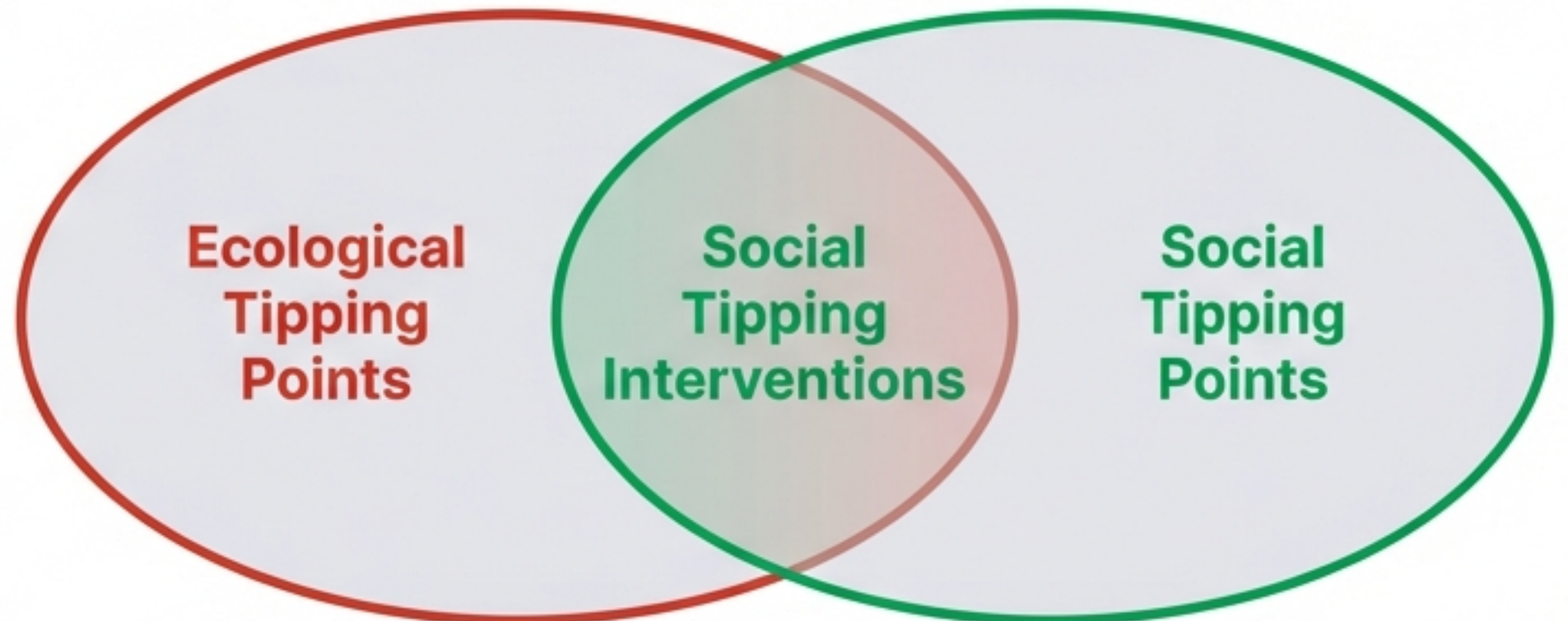
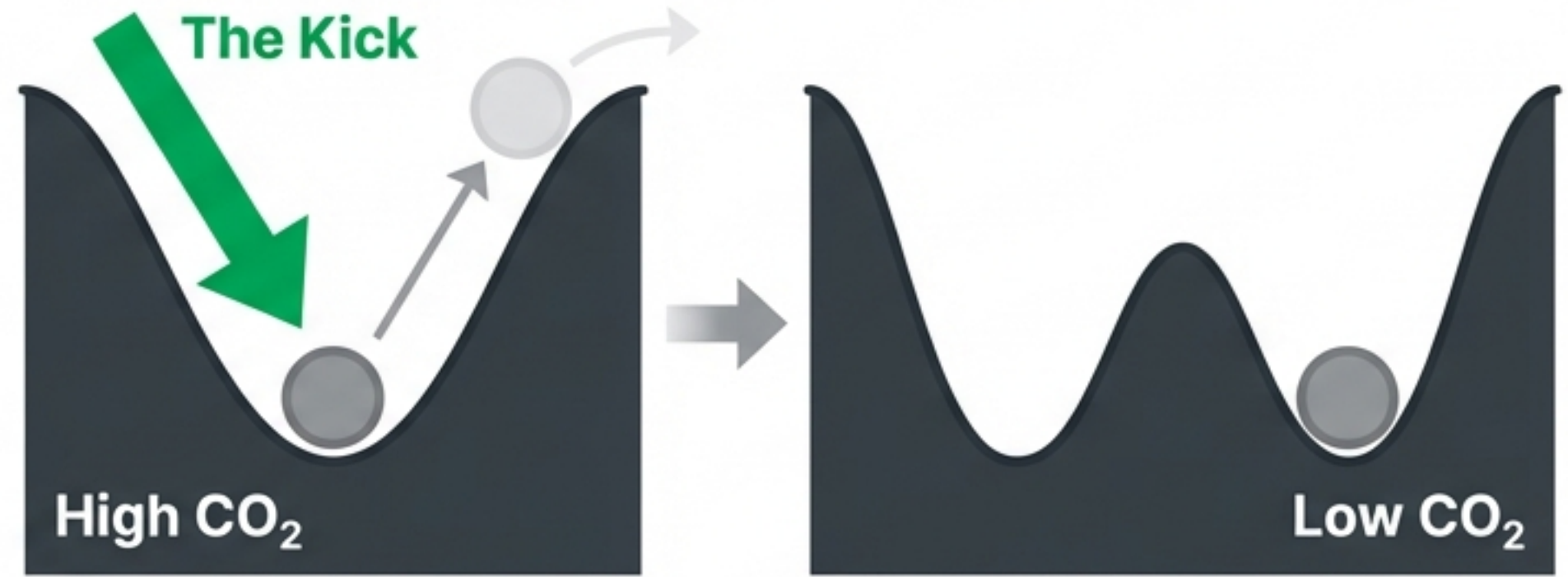
The rapid cessation of smoking and the abandonment of foot binding show how deeply entrenched norms can collapse quickly once a threshold is reached.

Designing 'The Kick': Social Tipping Interventions

We do not have to wait for tipping points to occur spontaneously. We can design “The Kick”—an external intervention that disrupts the status quo.

Concept Distinction

- **Ecological Tipping Points:** Often negative thresholds (e.g., Amazon dieback) that we must avoid.
- **Social Tipping Points:** Positive opportunities for rapid decarbonization that we must seize.



The Intervention Toolbox: Infrastructure, Information, and Incentives

Successful interventions reduce the friction for the majority to join the committed minority.



Infrastructure

Physical changes that make the new behaviour the “path of least resistance” (e.g., preferential lanes for EVs, bike infrastructure).



Information

Eco-labelling makes the “invisible” visible. It shifts consumer **behaviour** by highlighting environmental impact.

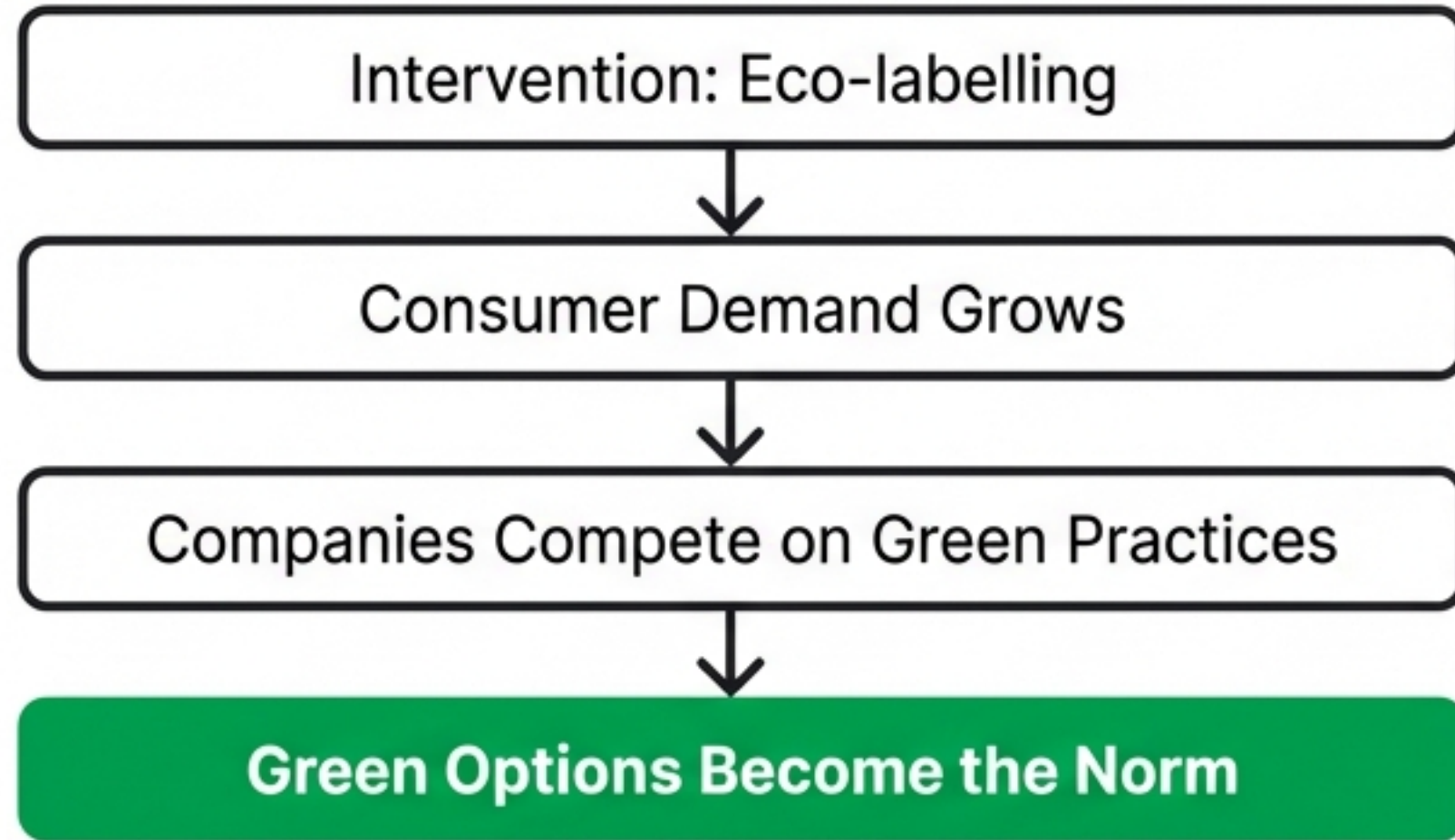


Incentives

Targeted subsidies that spark the initial adoption loop. Once the **behaviour normalizes**, the subsidy becomes less critical.

Sector Focus: Transforming Food and Transport

Food Systems



As demand for low-carbon food grows, companies compete on sustainability practices, shifting the market standard.

Transport Systems

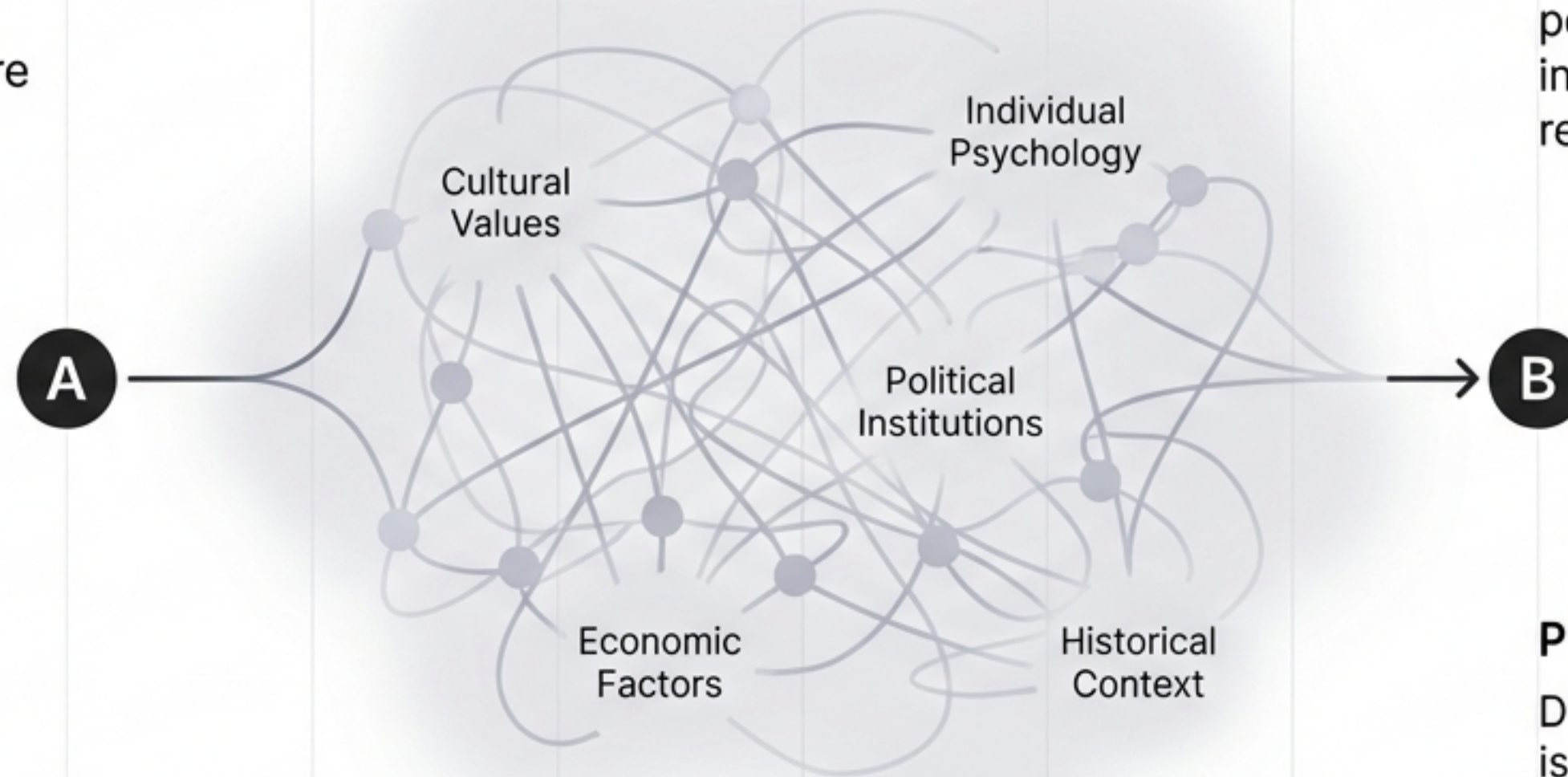
Intervention: Subsidies + Charging Infrastructure + Information.



Result: Rapid acceleration of **Electric Vehicle (EV)** adoption. It is not just about the engine; it is about social signaling. Even without a ban on combustion engines, the **social norm** can tip combustion engines, the **social norm** can tip toward EVs.

The Barrier of Complexity: Why Change is Unpredictable

Unlike natural systems (like fluid mechanics) where tipping points are tied to measurable physical factors, social systems are inherently messy.



Unmeasurable Variables

Systems are influenced by a volatile mix of cultural values, political institutions, and individual psychological responses.

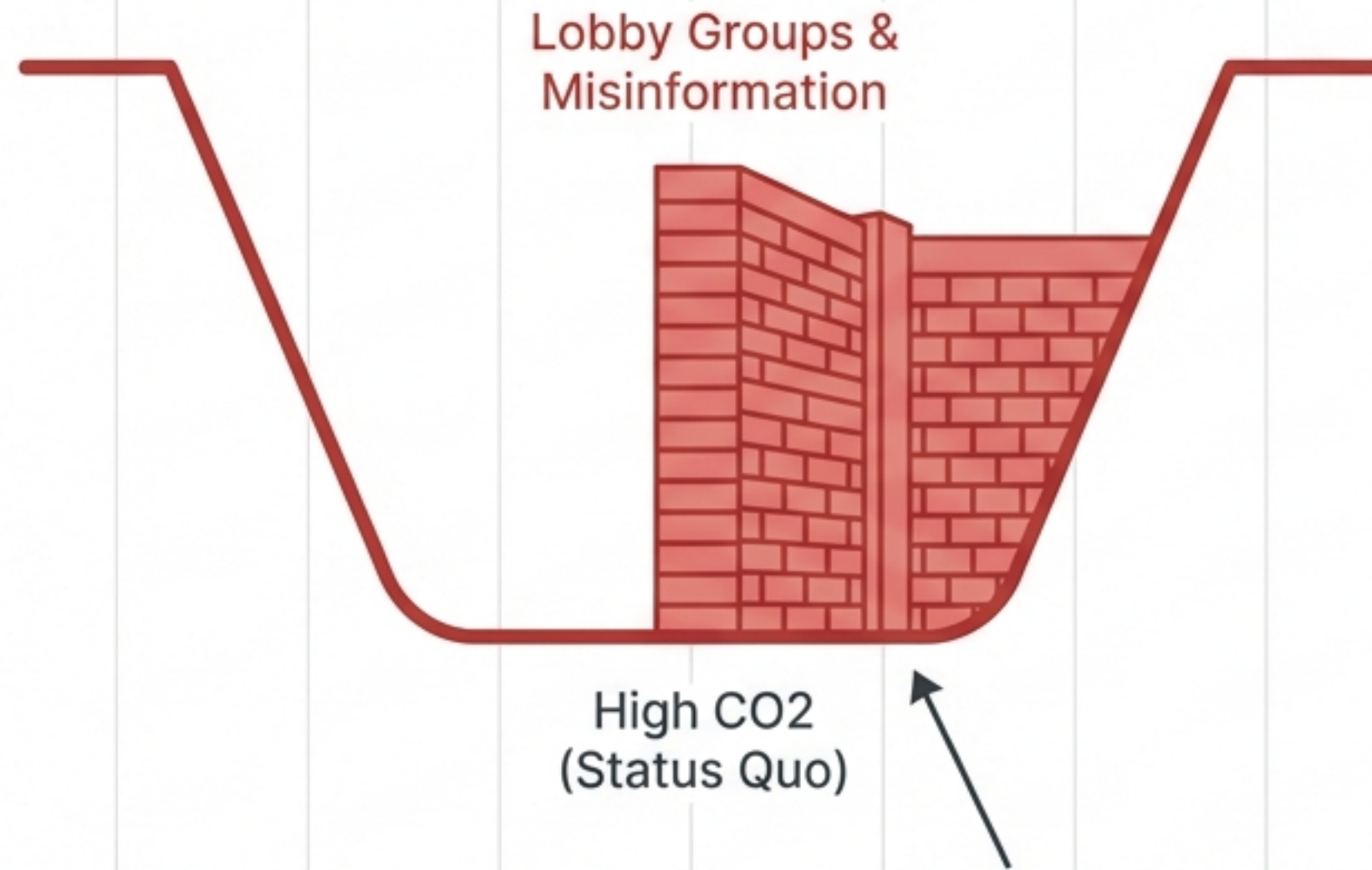
Prediction vs. Preparation

Disentangling every parameter is nearly impossible. It is hard to predict the exact moment a threshold will be reached, or to guarantee an intervention will work.

Risks: Polarization and Negative Tipping

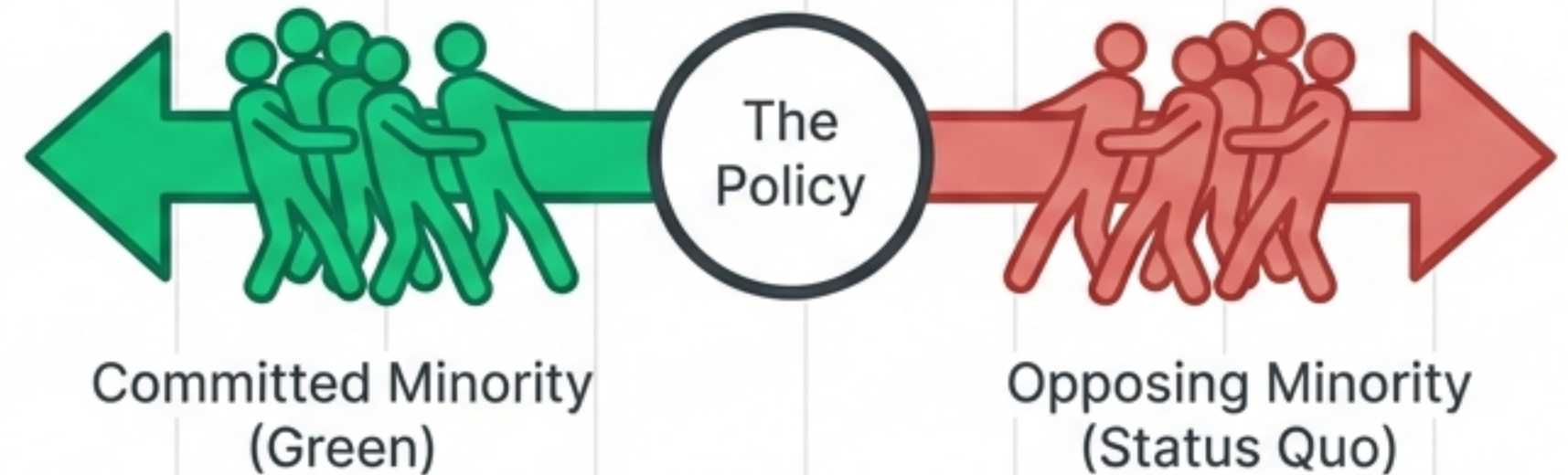
Social tipping is neutral; it can lead to a preferable equilibrium (decarbonization) or a socially undesirable one.

Deepening the Valley



Lobby groups use misinformation to "deepen the valley" of the status quo, increasing the barrier to change.

Polarization (The Tug of War)



Conflicting minorities (e.g., political polarization) can create a deadlock, preventing the system from tipping either way.

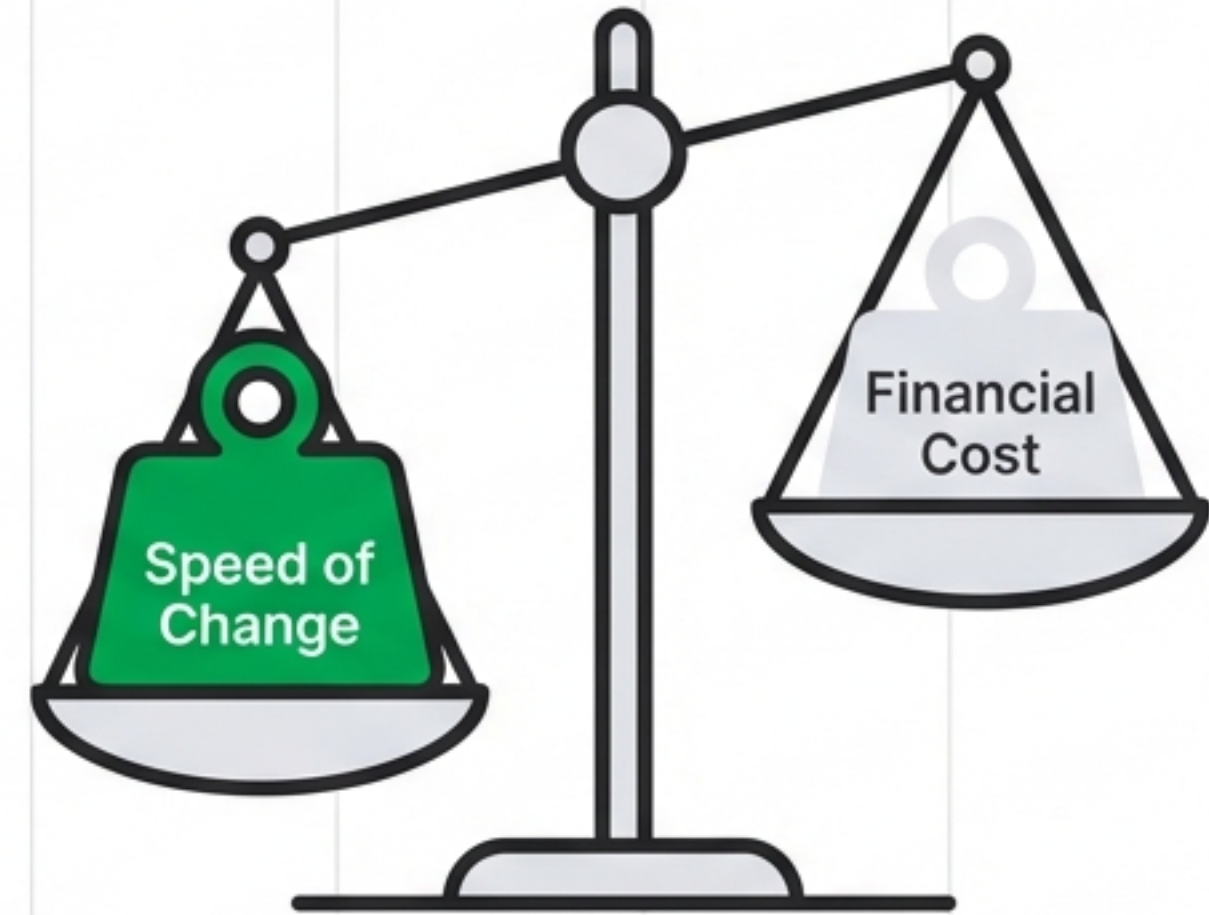
The Economic Reality: Cost and Effectiveness

While social tipping interventions often have higher public acceptability than traditional taxes, their economic efficiency is still under review.

Key Considerations

- **Resource Intensity:** Triggering a tipping point may require significant upfront financial resources to overcome resistance.
- **Uncertainty:** The cost-effectiveness of these interventions remains an open empirical question. Like all policy instruments, they face implementation hurdles and scaling difficulties.

Strategic View



Despite costs, the speed required to address climate change may make these interventions necessary complements to traditional policy.

The Strategic Imperative: Relentless Pressure

Change often appears gradual until it becomes sudden. The goal of leadership is not to predict the exact date of the tipping point, but to relentlessly build pressure until the threshold is crossed.

Empower the Minority

Support the 10-40% of early adopters.

Remove Friction

Design infrastructure that guides the “conformist majority” toward the new norm.

Design the Kick

Don't wait for the ball to roll; actively disrupt the equilibrium.

